

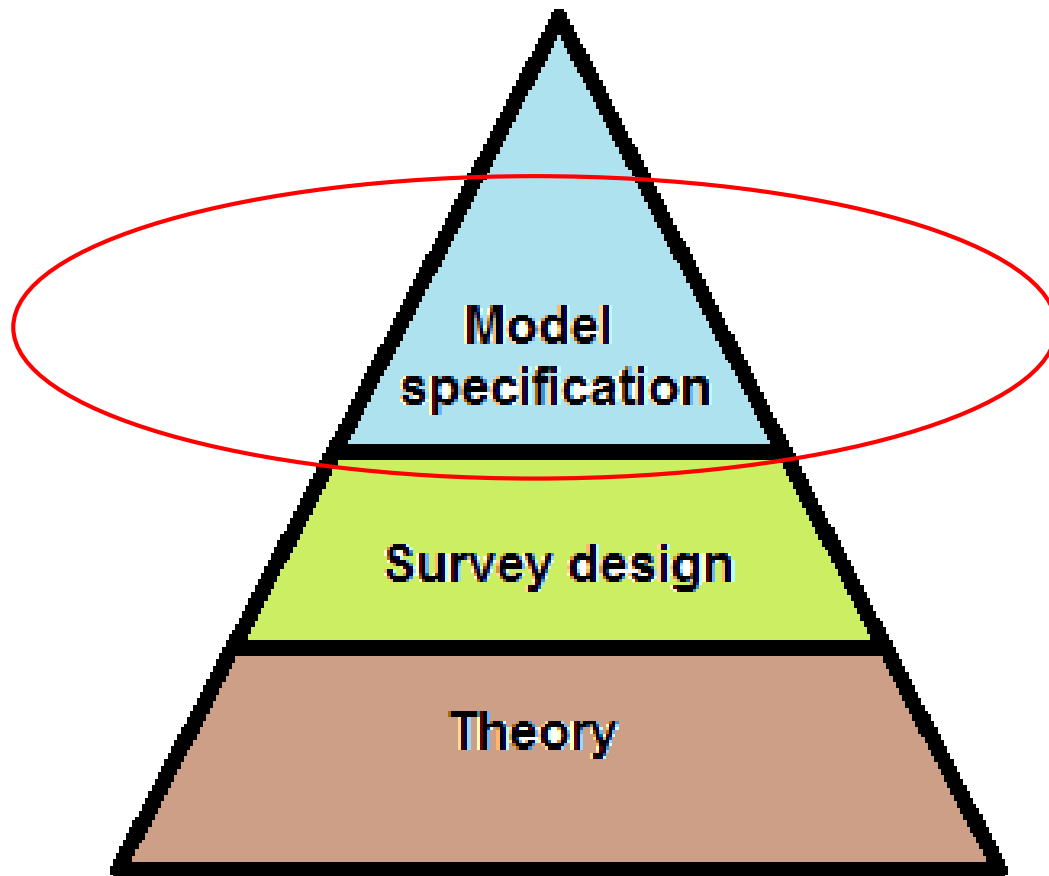
The Value of Travel Time

Insights from British and Danish national studies

hEART Conference – Leeds, September 2014



The Value of Travel Time Changes



The Value of Travel Time Changes

- Focus on national studies
 - ▣ **Value of travel time: key input for appraisal of projects**

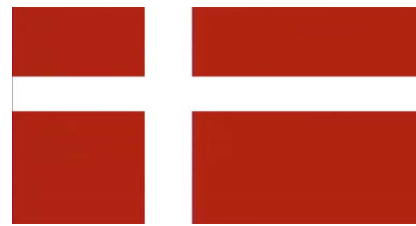


The Value of Travel Time Changes

- Numerous **issues** remain **unresolved**. Are we doing anything wrong? *How can we do things better?*

- Our contribution:

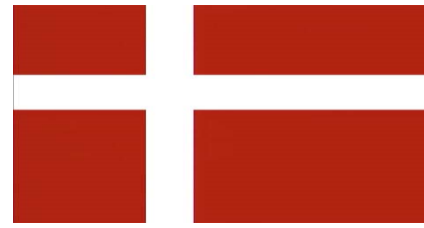
Comparative analysis of some elements **of two important national studies**.



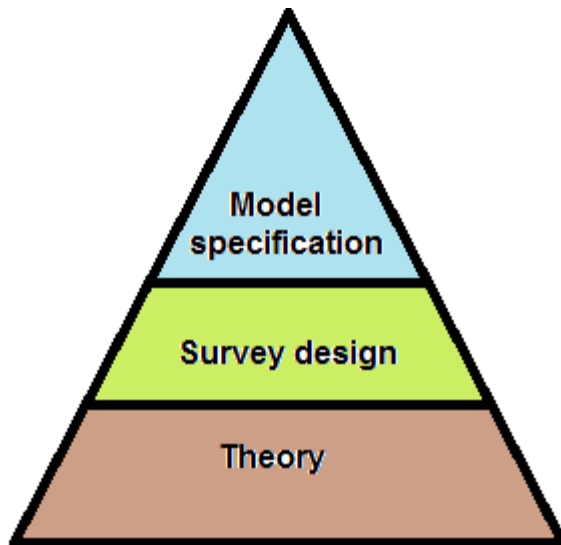
Two national studies on the Value of Time



Vs.



□ Why is this analysis interesting?



- 0. Travellers & context: **Different** (potentially)
- 1. Microeconomic theory
- 2. Design: **Identical***
- 3. Models: **Different**

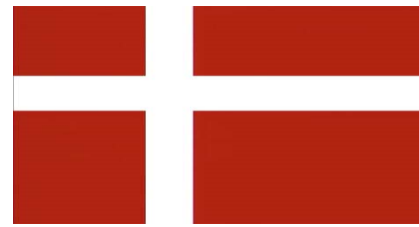
The Value of Travel Time Changes

- Two modelling approaches:

Random Utility



Random Valuation



The Value of Travel Time Changes

- Two modelling approaches:

Random Utility

$$y = 1\{V_1 > V_2 + \varepsilon\}$$

Random Valuation

Imagine a travel choice scenario such as follows:

	TRAVEL OPTION 1	TRAVEL OPTION 2
COST	£5	£10
TIME	60 min.	30 min.



Implicit valuation threshold:
$$\frac{|10 - 5|}{|30 - 60|} * 60 = 10 \text{ £/hour}$$

(known as: *Boundary VTTC*)

$$y = 1\{VTTC < BVTTC + \varepsilon\}$$

The Value of Travel Time Changes

- Two modelling approaches:

Random Utility

Random Valuation

Related to a particular type of Stated Choice data: 2 options & 2 attributes

This type of data is common in European national studies



The Value of Travel Time Changes

Random Utility

Random Valuation

□ Theoretical relationship?

The Value of Travel Time Changes

Random Utility

Random Valuation

- Theoretical relationship:
 - ▣ Both derived from **Microeconomic Theory**

$$U_i = (V_i, \varepsilon_i)$$

$$\text{where: } V_i = \beta_c c_i + \beta_t t_i$$

- ▣ $VTTC = \frac{\beta_t}{\beta_c}$

- ▣ **Difference: how is the ε_i introduced?**



The Value of Travel Time Changes

Random Utility

Random Valuation

- Theoretical relationship: Deterministic domain
 - ▣ If the slow option 1 is chosen, then $VTTC < BVTTC$:

$$\beta_t * t_1 + \beta_c * c_1 > \beta_t * t_2 + \beta_c * c_2$$

$$\beta_t * (t_1 - t_2) > -\beta_c * (c_1 - c_2)$$

$$\frac{\beta_t}{\beta_c} < -\frac{(c_1 - c_2)}{(t_1 - t_2)}$$

$$VTTC < BVTTC$$



The Value of Travel Time Changes

Random Utility

Random Valuation

□ Theoretical relationship: Stochastic domain

$$\begin{cases} \widetilde{U}_1 = \widehat{\beta}_t * t_1 + \widehat{\beta}_c * c_1 + \widetilde{\varepsilon}_1 \\ \widetilde{U}_2 = \widehat{\beta}_t * t_2 + \widehat{\beta}_c * c_2 + \widetilde{\varepsilon}_2 \end{cases} \quad (1) \quad \longleftrightarrow \quad \begin{cases} U_1 = \mu * BVTTC + \varepsilon_1 \\ U_2 = \mu * VTTC + \varepsilon_2 \end{cases} \quad (2)$$

Where:

ε_i are i.i.d.

μ : scale parameter

$\widehat{\beta}_t = \mu\beta_t$

$\widehat{\beta}_c = \mu\beta_c$

□ Rearrange (2):

$$\begin{cases} U_1 = 0 + \varepsilon_1 \\ U_2 = \mu * VTTC - \mu * BVTTC + \varepsilon_2 \end{cases} \quad (3)$$

□ Multiply (3) by β_c and Δt :

$$\begin{cases} \Delta t \beta_c U_1 = 0 + \Delta t * \beta_c * \varepsilon_1 \\ \Delta t \beta_c U_2 = \underbrace{\Delta t * \beta_c * \mu * VTTC}_{\mu * \beta_t * \Delta t} - \underbrace{\Delta t * \beta_c * \mu * BVTTC}_{\mu * \beta_c * \Delta c} + \Delta t * \beta_c * \varepsilon_2 \end{cases}$$



The Value of Travel Time Changes

Random Utility

Random Valuation

- Theoretical relationship: Stochastic domain

$$\begin{cases} \bar{U}_1 = \hat{\beta}_t * t_1 + \hat{\beta}_c * c_1 + \tilde{\varepsilon}_1 \\ \bar{U}_2 = \hat{\beta}_t * t_2 + \hat{\beta}_c * c_2 + \tilde{\varepsilon}_2 \end{cases} \iff \begin{cases} U_1 = \mu * \text{BVTTC} + \varepsilon_1 \\ U_2 = \mu * \text{VTTC} + \varepsilon_2 \end{cases}$$

$$\tilde{\varepsilon}_i = \Delta t * \beta_c * \varepsilon_i$$

Difference: particular
form of heteroskedastic
errors

The Value of Travel Time Changes

Random Utility

Random Valuation

□ Empirical comparison?



The Value of Travel Time Changes

Random Utility

Random Valuation

- Empirical comparison?
 - ▣ 1) Linear
 - ▣ 2) in Logarithms
 - 3) + observed heterogeneity
 - 4) + random heterogeneity



The Value of Travel Time Changes

Random Utility

$$1 \quad \begin{cases} U_1 = \beta_c \left(\frac{\beta_t}{\beta_c} * t_1 + c_1 \right) + \varepsilon_1 \\ U_2 = \beta_c \left(\frac{\beta_t}{\beta_c} * t_2 + c_2 \right) + \varepsilon_2 \end{cases}$$

$$2 \quad \begin{cases} U_1' = \mu * \ln(VTTC * t_1 + c_1) + \varepsilon_1' \\ U_2' = \mu * \ln(VTTC * t_2 + c_2) + \varepsilon_2' \end{cases}$$

$$3 \quad VTTC = e^{\beta_0 + \beta_{BC} \ln\left(\frac{C}{C_0}\right) + \beta_{BT} \ln\left(\frac{T}{T_0}\right) + \beta_I \ln\left(\frac{I}{I_0}\right)} = \beta_0 * \left(\frac{C}{C_0}\right)^{\beta_{BC}} \left(\frac{T}{T_0}\right)^{\beta_{BT}} \left(\frac{I}{I_0}\right)^{\beta_I}$$

$$4 \quad VTTC = e^{\beta_0 + \beta_{BC} \ln\left(\frac{C}{C_0}\right) + \beta_{BT} \ln\left(\frac{T}{T_0}\right) + \beta_I \ln\left(\frac{I}{I_0}\right) + u}$$

Random Valuation

$$\begin{cases} U_1 = \mu * BVTTC + \varepsilon_1 \\ U_2 = \mu * VTTC + \varepsilon_2 \end{cases}$$

$$; VTTC = \frac{\beta_t}{\beta_c} = \beta_0$$

$$\begin{cases} U_1' = \mu * \ln(BVTTC) + \varepsilon_1' \\ U_2' = \mu * \ln(VTTC) + \varepsilon_2' \end{cases}$$

Two national studies on the Value of Time

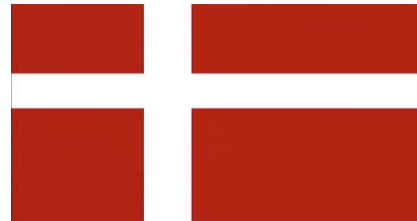
□ Design & dataset in each country:

- *Common SC design, implemented slightly differently:*

- 1. Travellers & context:
Different
- 2. Design:
Identical*



	TRAVEL OPTION 1	TRAVEL OPTION 2
COST change (Δc)	100 pence	0
TIME change (Δt)	-15 min.	0



	TRAVEL OPTION 1	TRAVEL OPTION 2
COST	400 pence	300 pence
TIME	30 min.	45 min.

(Illustrative values in £, instead of DKK, for comparison)

The Value of Travel Time Changes

	1. Linear				2. Logarithms			
	RU		RV		RU		RV	
	Est.	t-test	Est.	t-test	Est.	t-test	Est.	t-test
β_c	1	na	na	na	1	na	na	na
β_0	4.89	18.64	3.22	11.76	3.71	22.38	2.75	23.71
μ	-0.0138	-20.81	0.115	24.03	-6.42	-23.12	0.79	33.15
VTTC pence/min	4.89		3.22		3.71		2.75	
Obs.	10598		10598		10598		10598	
Parameters	2		2		2		2	
Null LL	-7345.974		-7345.974		-7345.974		-7345.974	
Final LL	-6746.152		-6570.224		-6690.042		-6465.961	
Adj. Rho ²	0.081		0.105		0.089		0.120	



The Value of Travel Time Changes

	3. Logarithms + Covariates				4. Log + Covariates + Random Het.			
	RU		RV		RU		RV	
	Est.	t-test	Est.	t-test	Est.	t-test	Est.	t-test
β_c	1	na	na	na	1	na	na	na
β_0	1.70	30.23	1.30	28.25	1.58	28.64	1.29	28.11
μ	7.39	25.26	0.859	34.24	11.5	24.06	1.09	33.00
β_{BC}	0.470	8.78	0.431	7.57	0.431	7.45	0.428	25.29
β_{BT}	-0.362	-4.81	-0.196	-2.68	-0.279	-3.50	-0.189	-2.61
β_I	0.273	5.13	0.411	8.06	0.344	6.40	0.382	7.77
σ	na	na	na	na	1.07	21.22	1.11	25.29
VTTc pence/min	5.47		3.67		8.61		6.72	
Obs.	10598		10598		10598		10598	
Parameters	5		5		6		6	
Null LL	-7345.974		-7345.974		-7345.974		-7345.974	
Final LL	-6607.502		-6300.028		-6306.561		-5910.137	
Adj. Rho ²	0.100		0.142		0.141		0.195	



The Value of Travel Time Changes

	1. Linear				2. Logarithms			
	RU		RV		RU		RV	
	Est.	t-test	Est.	t-test	Est.	t-test	Est.	t-test
β_c	1	na	na	na	1	na	na	na
β_0	63.3	14.6	20.5	11.9	31	13.3	31.1	23.91
μ	-0.00058	-14.5	0.0169	28.7	-3.14	-18.47	0.711	35.36
VTTC DKK/hour	37.95		20.5		18.6		18.66	
Obs.	17020		17020		17020		17020	
Parameters	2		2		2		2	
Null LL	-11797.4		-11797.4		-11797.4		-11797.4	
Final LL	-11378.7		-10807.6		-10922.3		-10763.2	
Adj. Rho ²	0.035		0.084		0.074		0.087	

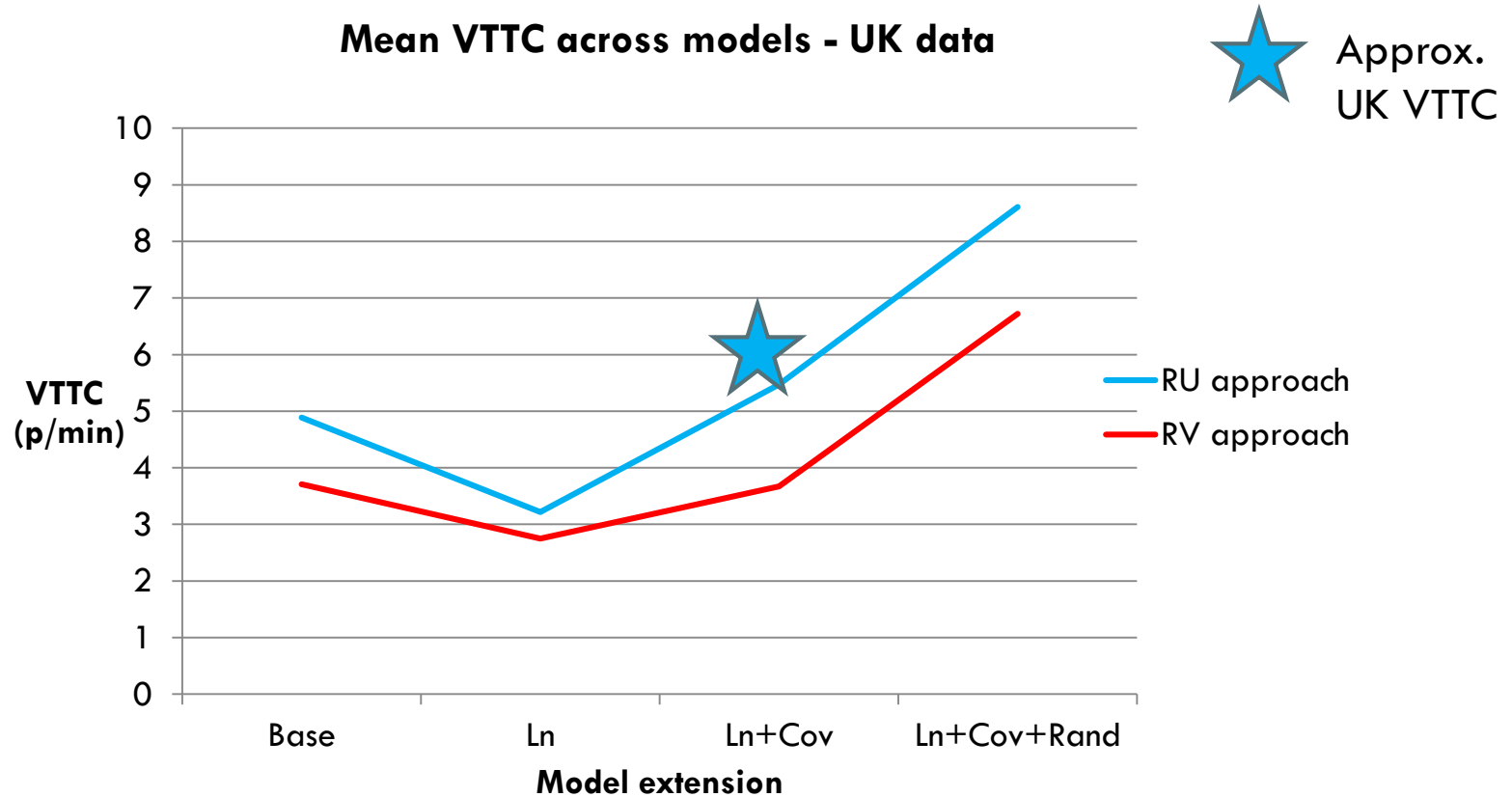


The Value of Travel Time Changes

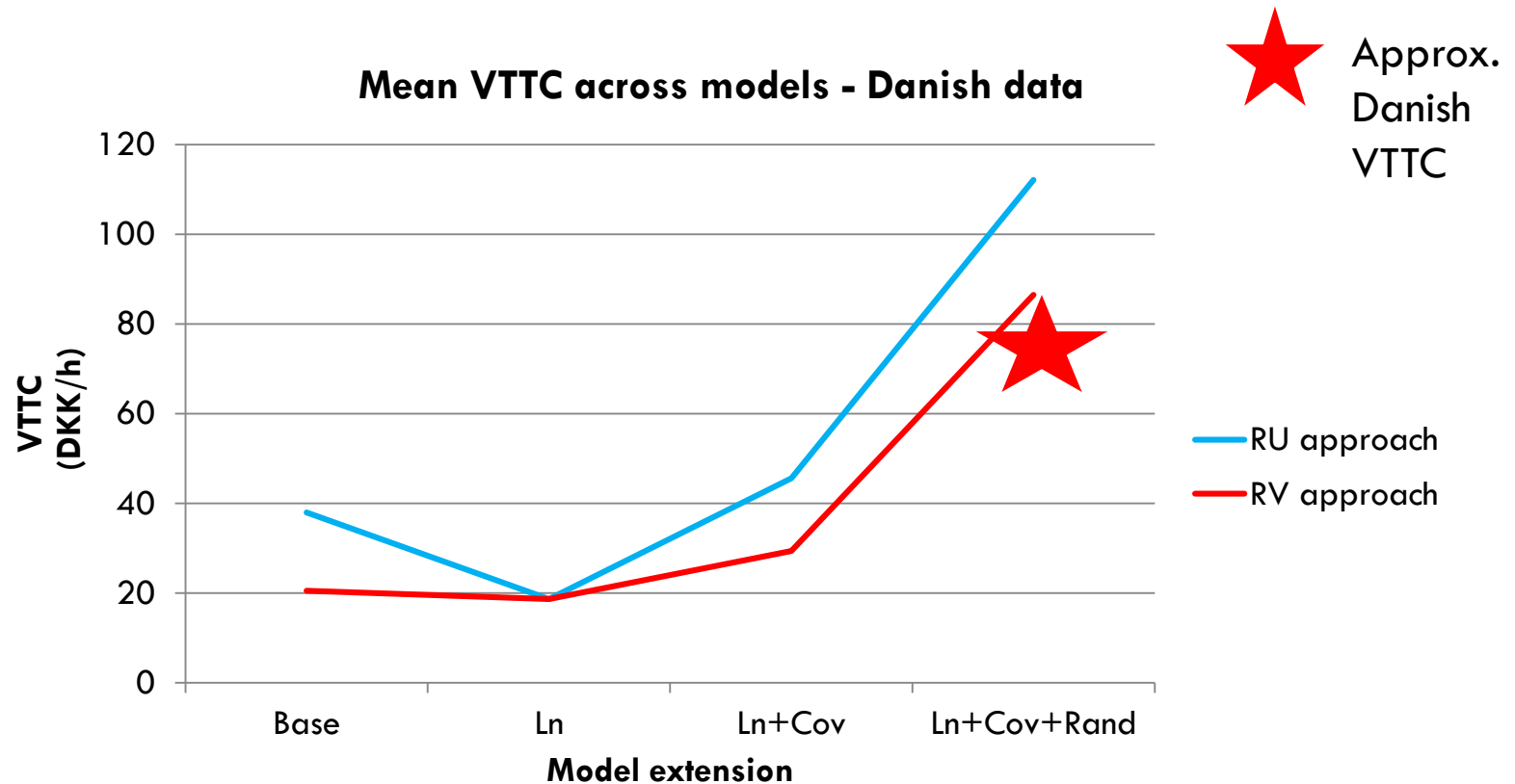
	3. Logarithms + Covariates				4. Log + Covariates + Random Het.			
	RU		RV		RU		RV	
	Est.	t-test	Est.	t-test	Est.	t-test	Est.	t-test
β_c	1	na	na	na	1	na	na	na
β_0	4.33	62.25	3.89	87.25	4.12	68.27	3.89	84.77
μ	4.41	20.37	0.768	36.3	10.3	24.4	1.06	34.84
β_{BC}	0.571	6.51	0.701	9.44	0.581	7.00	0.705	9.23
β_{BT}	-0.48	-3.87	-0.643	-6.06	-0.451	-3.67	-0.633	-5.77
β_I	0.501	6.63	0.638	9.76	0.611	8.47	0.633	9.74
σ	na	na	na	na	1.49	26.8	1.47	30.21
VTTC DKK/hour	45.57		29.35		112.08		86.45	
Obs.	17020		17020		17020		17020	
Parameters	5		5		6		6	
Null LL	-11797.4		-11797.4		-11797.4		-11797.4	
Final LL	-10748.8		-10313.8		-9690.48		-9185.81	
Adj. Rho ²	0.088		0.125		0.178		0.221	



The Value of Travel Time Changes



The Value of Travel Time Changes



The Value of Travel Time Changes

- Some results:
 - **The choice of approach matters.**
 - Theoretically, none of the approaches is preferred.
 - Empirically:
 - **RandomValuation** > **RandomUtility** (model fit)
 - Lower *Value of Time* using **RV approach**
 - Similar pattern across models in UK & Denmark



The Value of Travel Time Changes

- Some issues and recommendations:
 - **Risk of significant biases** on VTTC depending on the form of error heteroskedasticity selected.
 - **RV preferred** if feasible: e.g. with simple 2options&2attributes experiments.
 - **RU** approach: likely to have heteroskedastic errors that **need correction**.
 - Test both linear and logs specifications.



The Value of Travel Time Changes

- Some questions:
 - **Why is the VTTC always lower** with the RV approach?
 - **Validity of** results using **Random Utility** approach? Always worse model fit.
 - **Why are individuals' preferences so similar in two different countries?** What is the role of stated choice **design** on results?

Thanks

References

The archived submission did not contain a separately labelled references section.